

GEOVARIABILITY AND BIODEGRADABILITY OF THE WATER-EXTRACTABLE ORGANIC MATERIAL IN AN AGRICULTURAL SOIL

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Summary—An analysis of the quantity, biodegradability, three-dimensional distribution and u.v. absorption at 254 nm of the water-extractable organic carbon (WEOC) from an agricultural loess soil was made. In addition, the total organic carbon (TOC) in the samples was quantified. The following results were obtained. The average concentration ($9.3 \mu\text{g g}^{-1}$ DW) of the WEOC was fairly constant to a depth of 50 cm, which was approximately the boundary between the B and C horizons. Below 50 cm it again remained constant at significantly lower amounts ($6.5 \mu\text{g g}^{-1}$ DW) down to 100 cm. The WEOC consisted of two fractions: one that was not biodegradable with an average concentration of $3.9 \mu\text{g g}^{-1}$ DW and one that was 85% degradable under the conditions used. This was independent of depth. The WEOC in the Ap horizon absorbed more u.v. radiation than the WEOC from the deeper horizons. This was especially the case for the refractory material. An analysis of the data with semivariograms indicated that the WEOC concentrations in the upper 50 cm were random while the ones below this depth were isotropic, being dependent on the direction of tillage. This may have been the result of preferential flow, resulting from root channels. Another possibility is organic solute enrichment below the crop rows due to the water movement into the plants. TOC decreased sharply below 25 cm depth and tended to decline further with depth. The amount of the WEOC relative to TOC increased with depth. There was no tight statistical covariation between WEOC and TOC.

INTRODUCTION

In soil chemistry, chemical compounds are often classified by some physico-chemical criterion, usually solubility. Classical examples of this are the humic and fulvic acids. Fulvic acids, for example, are water soluble after extraction from soil but are not water extractable. The material analyzed in this investigation was water extractable and is the source, when sufficient water is present, of the dissolved organic carbon or material, which is found in leachate (the mobile organic material). The position of this water-extractable organic material (WEOM) in a conceptual model of the organic matter in a soil is shown in Fig. 1. A classification of the organic material in soils based on water solubility makes ecological sense, since organic material is transferred from one ecological compartment to another as a result of water movement.

The ecological significance of the WEOM has been investigated in several areas. To the extent that it is composed of organic acids, it can help weather rock (Antweiler and Drever, 1983), mobilize metals (Krug and Isaacson, 1984; Chen and Stevenson, 1986) and disperse clays. This latter effect can interfere with the formation of aggregates (Oades, 1984). On the other hand the WEOM has also been shown to stabilize aggregates, presumably as a function of its carbohydrate content (Chaney and Swift, 1984; Cheshire *et al.*, 1984; Oades, 1984). The mobilization or cotransport of herbicides by the WEOM has also been reported (Madhun *et al.*, 1986). It is difficult to quantify the extent that substrates need to be water

soluble before they are available to the micro heterotrophs in soil, but McGill *et al.* (1986) assume that this is generally the case. The WEOM may also play a significant role in the nitrogen cycle. Burford and Bremner (1975) found a close connection between the denitrification capacity of a soil and the content of the WEOM. Cassman and Munns (1980) found that up to 40% of the mineralizable nitrogen in soil was located below a depth of 36 cm. It can be theorized that some of it was transported there by the WEOM.

Our purpose was (1) to analyze the geovariability, both horizontally and vertically, of the WEOM in a typical agricultural soil; (2) to obtain an estimate of the extent that the WEOM can be mineralized by microbes; and (3) to see if there is any apparent relationship between the WEOM and the TOC. In addition, the WEOM was roughly characterized by its ability to absorb in the u.v.

MATERIALS AND METHODS

Location and sampling

Samples were taken in 1990 from a loess soil in the neighborhood of Regensburg. The horizons were: 0–30 cm, Ap; 30–45 cm Bt; 45–70 cm, Cr; and 70–100, Bwb. A 25×25 m square grid was laid out. A core of 100 cm depth was taken at every 5 m with an auger to give samples representative of the soil at 0–25, 25–50, 50–75 and 75–100 cm depth, a total of 144 samples in all. Sugar beet at the four leaf stage were present on the field at the time of sampling. Previously, the field was fallow after a winter wheat crop

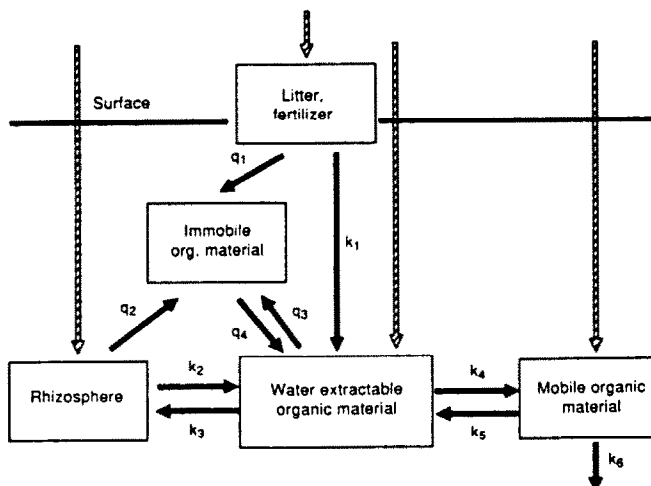


Fig. 1. A conceptual model of the flow of organic material in a soil. The hatched arrows refer to precipitation. The rates k_i are most likely directly coupled to precipitation rates, while the q_i are not.

had been harvested in August 1989. No organic fertilizers had been used.

Extraction and analysis

Soil weighing 50 g was extracted by gently stirring for about 1 min with 100 ml of a 10 mM CaCl_2 solution at room temperature (22°C). Simple comparison tests indicated that the length and vigor of stirring was not critical. More important was that the extraction temperature be held constant (R. Schierl, personal communication; A. Zsolnay, unpublished data). The samples were allowed to settle for 10 min and were then decanted through a filter with a pore size of 1.0 μm . Polycarbonate (Nuclepore) filters were found to have the least amount of bleed and adsorption. A pore size of 1.0 μm rather than the usual 0.45 μm was used, since non-sterile conditions were desired for the microbial decomposition phase of the experiment. There were no significant quantitative differences in the concentration of the WEOM obtained by filtration with 1.0 or 0.4 μm filters. Half of the filtrate (ca 30 ml) was stored in the dark in closed 150 ml brown bottles at room temperature for 3 months. Mineralization of readily-decomposable organic material has been found to go to completion within 3 months under such conditions (Zsolnay, 1975). At the end of this time, the incubated filtrate was analyzed in the same manner as the non-incubated half. The water extractable organic C (WEOC) was measured with a Heraeus liquid TOC, based on selective wet oxidation under u.v. radiation followed by i.r. quantification of the CO_2 released. The coefficient of variation was 3.5%, which included errors due to sample heterogeneity, sample extraction and analysis.

A portion of soil from each sample was dried in an oven at 105°C to obtain the dry weight (DW). The interstitial water contents of the soil samples (250 $\mu\text{l g}^{-1}$ DW on an average) were taken into consideration in calculating the WEOC concentrations and the u.v. absorptivity. Total organic C for the soil sample (TOC) was measured with a Carlo Erba NA 1500 Nitrogen Analyzer, based on catalytic combustion and gas solid chromatography. Carbonate content

was corrected for. The coefficient of variation for the TOC procedure was 0.5%.

Ultraviolet absorption was measured at 254 nm with a Hitachi UV-2000 Spectrophotometer. The absorption of nitrate and nitrite caused no interference at this wavelength. pH of the soils in a 10 mM CaCl_2 solution was measured with a glass electrode. The average pH was 7.6. The semivariogram analyses were done with GEO-EAS (Geostatistical Environmental Assessment Software) from the United States Environmental Protection Agency (Englund and Sparks, 1988).

RESULTS

The WEOC amounts per dry weight soils and per volume interstitial water are given in Table 1 and in Fig. 2 as a function of depth. The notched box plots provide a robust, non-parametric test for comparisons (McGill *et al.*, 1978). The central line in the boxes is the median, and the notches cover the 95% confidence interval of the median. Samples can be considered to be significantly quantitatively different when their notches do not overlap. The hinges are at the 25 and 75% quartiles, while the "whiskers" encompass all data within an extension of the quartiles by 1.5 times the interquartile range. Values larger or smaller than this are shown with a (+) and would most likely be outliers in a parametric analysis of the data. The u.v. absorptivity of the WEOC at 254 nm is given in Table 1 and in Fig. 3.

Table 1. Average values for water-extractable (WEOC) and total organic carbon (TOC) obtained as a function of soil depth

Depth (m)	WEOC* ($\mu\text{g g}^{-1}$)	WEOC† (mg l^{-1})	TOC* (mg g^{-1})	WEOC/TOC (%)	u.v.‡ ($\text{l mg}^{-1} \text{m}^{-1}$)
0-25	8.9	35.6	11.8	0.08	1.44
25-50	9.7	37.7	5.1	0.19	0.79
50-75	6.6	26.7	2.2	0.30	0.84
75-100	6.3	27.4	1.6	0.39	0.71

*Based on soil oven dry weight.

†Based on interstitial water.

‡Absorptivity of the WEOC at 254 nm.

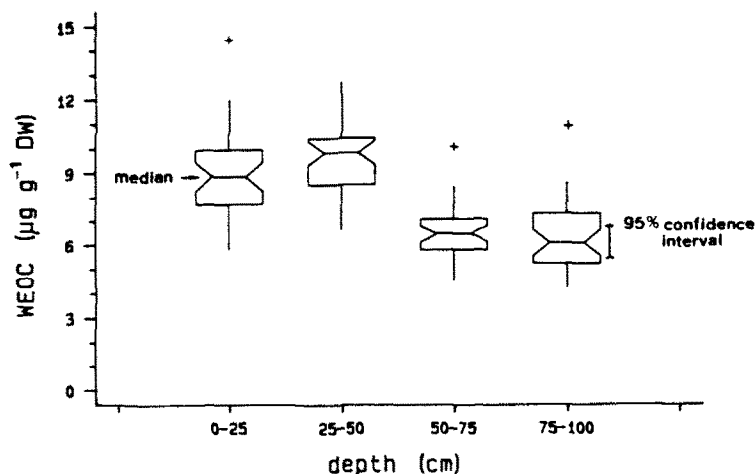


Fig. 2. Notched box chart of the concentration of water-extractable organic carbon (WEOC) per unit oven dry weight soil (DW) as a function of depth. Thirty-six samples were taken at each depth (cf. text for chart conventions).

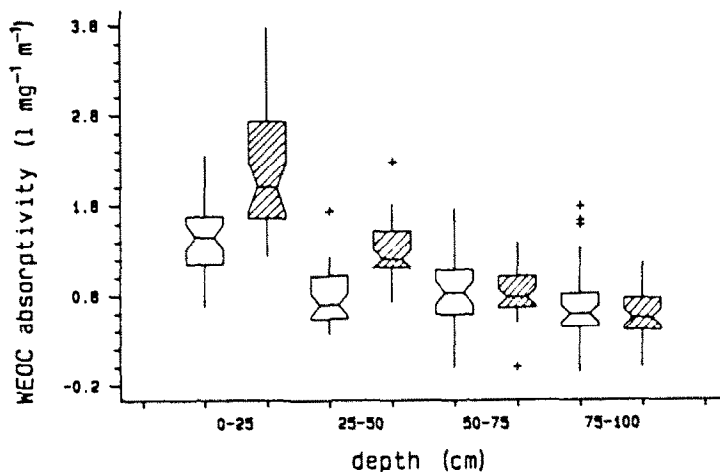


Fig. 3. Notched box chart of the u.v. absorbivity at 254 nm of the water-extractable organic carbon (WEOC) as a function of depth. The open boxes refer to the original material, the hatched ones to material, which was refractory to microbial mineralization. Each box represents 36 samples (cf. text for chart conventions).

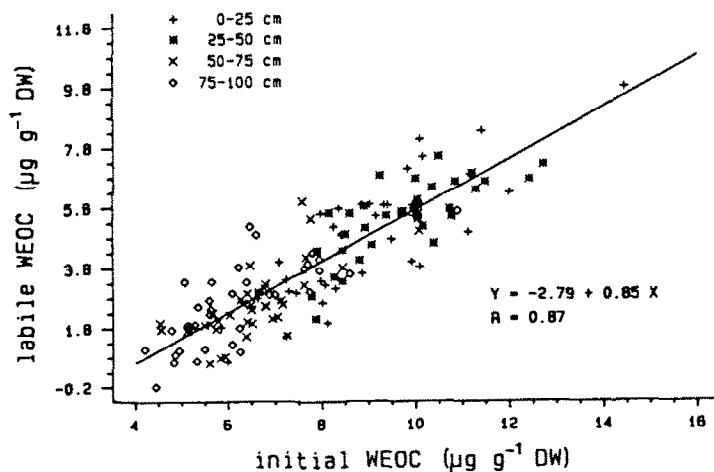


Fig. 4. Plot of a linear regression analysis of labile water-extractable organic carbon (WEOC) as a function of the total or initial WEOC. Concentrations are per unit oven dry weight (DW) soil. The symbols indicate at which depth the samples were taken.

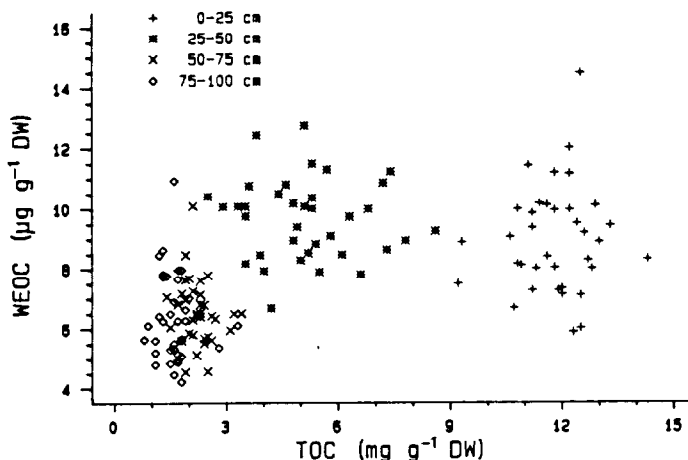


Fig. 5. Scatter plot of the concentrations of water-extractable organic carbon (WEOC) vs total organic carbon (TOC) per unit oven dry weight soil. The symbols indicate at which depth the samples were taken.

The WEOC remaining after the 3 months incubation was considered to be refractory and was on an average $3.9 \mu\text{g g}^{-1}$ DW. The u.v. absorptivity of this material at 254 nm is also shown in Fig. 3. Labile WEOC amounts were obtained by subtracting the refractory material from the total WEOC. The results of a simple linear regression analysis of the dependence of the labile WEOC on the total or initial WEOC are shown in Fig. 4. Although regression analysis of this type are only strictly permissible when the independent values are free from error, it

was thought to be justifiable here, since the range of the independent values was sufficiently greater than the error of any one independent values. With the resulting statistical model 76% of the variation of the labile WEOC could be explained as a linear function of the total or initial WEOC.

TOC concentrations at the different depths are given in Table 1. The quantitative relationship between TOC and WEOC is shown in Fig. 5 and Table 1.

The semivariograms for the WEOC at two different depths are given in Fig. 6. They are shown parallel and perpendicular to the direction of tillage. A fairly large tolerance of 20° was used, because of the relatively small number of samples. Also, because of the small sample size, the results are rather more qualitative than quantitative. Only values resulting from at least 20 pairs are plotted. A semivariogram is a plot of one-half the variance of pairs of measurements of a parameter, in this case WEOC, against the distances separating the pairs (Webster, 1977; England and Sparks, 1988). The distribution of a parameter shows structure when the samples, which are physically closer to each other, are more similar than those which are further apart. In Fig. 6, this is only the case of the WEOC from 75 to 100 cm depth, which was parallel to the tillage. If the existence of this structure is dependent on a geographical direction as well as distance, the parameter is isotropic. This is again the case for the WEOC from 75 to 100 cm depth. The semivariograms for the WEOC from 50 to 75 cm depth were similar to those of the material from 75 to 100 cm, while the WEOC distribution from 25 to 50 cm depth showed no structure and in this way was similar to the material from 0 to 25 cm. Semivariograms parallel and perpendicular to tillage for the TOC were also plotted for all the sample depths, but the results in all cases showed no apparent structure.

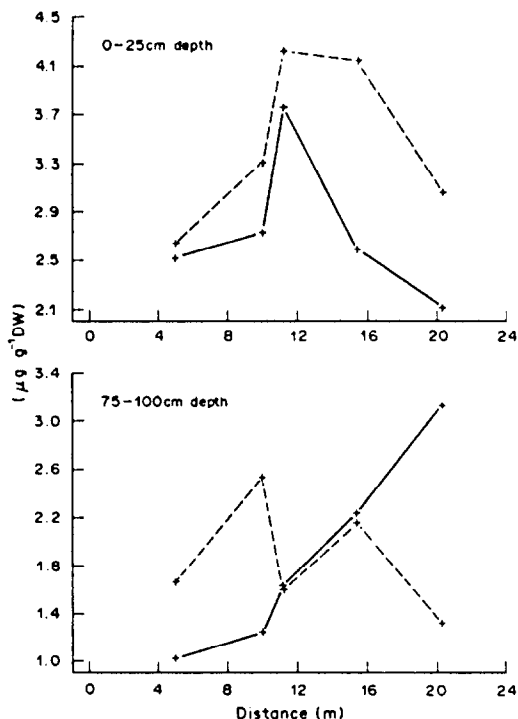


Fig. 6. Semivariograms of the concentration of the water-extractable organic material at two different depths and with two different orientations. The solid line was determined parallel while the dashed line was determined perpendicular to tillage. The angle of tolerance was 20° .

DISCUSSION

Average WEOC concentrations in Ap horizons of $30 \mu\text{g g}^{-1}$ DW (McGill *et al.*, 1986) and $40 \mu\text{g g}^{-1}$ DW (Burford and Bremner, 1975) have been

reported. These are somewhat higher than the concentration found here in the upper 25 cm (Table 1 and Fig. 2). Most values in the literature deal with leachate (the mobile organic matter in Fig. 1). Reported values for mineral soil leachate range from 6 mg l⁻¹ in a forest B horizon (Vance and David, 1989) to 50 mg l⁻¹ in a mountain meadow (Antweiler and Drever, 1983). The WEOC concentrations reported here for interstitial water (Table 1) fit in well, even though they are presumably larger than what is present *in situ* or in a leachate, since all water-extractable material was obtained, and not only the organic matter, which was dissolved in the interstitial water of the sampled soil. The WEOC amounts measured here are quite small compared to the TOC (Table 1). However, the turnover of the WEOC is much greater than that of most of the immobile material. Using the assumptions of McGill *et al.* (1986) a flow of 330 g WEOC m⁻² a⁻¹ with an estimated pool of 4600 g m⁻² immobile organic C can be calculated for the Ap from the samples analyzed here.

The general sources of the WEOC are given in Fig. 1: (1) near surface litter and fertilizer (Guggenberger *et al.*, 1989; especially when the fauna is active, Huhta *et al.*, 1988); (2) immobile organic matter; (3) the rhizosphere (through exudates, Beck and Gilmour, 1983; as well as sloughed material, Griffin *et al.*, 1976); and (4) the mobile organic material moving downwards from the shallower horizons. Each of these sources is intimately associated with a microbial population, which may markedly influence the quantity and nature of the WEOC produced there (Barber and Martin, 1976). The samples analyzed here were from a field with no fresh litter or organic fertilizers and living rhizosphere activity was still minimal. Therefore, the only major source of WEOC in the Ap could have been the immobile organic material, while in the deeper horizons augmentation through the mobile organic material coming from the upper horizons could also have played a role. Figure 5 and Table 1, however, show that the relationship between WEOC and TOC (mainly immobile organic C) is not a simple one at any of the depths sampled. When one analyzes the data of Burford and Bremner (1975), which were obtained from 16 different surface agricultural soils, also no covariation ($R = -0.25$) can be seen between WEOC and TOC. A reasonable conclusion is that only a portion of the immobile material is capable of being transferred into WEOM. This is again confirmed by an analysis of the data from Burford and Bremner (1975), which show a close covariation between WEOC and total *mineralizable* organic C ($R = 0.96$).

Figure 4 shows that the WEOC itself is also composed of two different pools of organic compounds. One pool contained material, which could be 85% mineralized, while the other pool was refractory containing a relatively constant amount (3.9 µg g⁻¹ DW on average). The calculated amount from the regression equation is 3.3 µg g⁻¹. These findings were essentially independent of depth. Figure 3 shows that the refractory WEOC in the upper 50 cm has a greater ability to absorb u.v. radiation than the total WEOC. This can be explained by the presence of lignin and other difficult-to-break-down substituted

phenolic compounds near the surface (Kuiters and Sarink, 1985).

The WEOC concentration from 25 to 50 cm (roughly the Bt horizon) is, on an average, 9% larger than that of the material in the Ap horizon, even though the TOC concentration has declined by 57%. Therefore, some, if not most, of the WEOC was transported here from the Ap via the mobile organic material. However, there appears to be a definite boundary to this transport at about 50 cm depth, since there is a significant drop in the WEOC below this depth (Fig. 2). This may have been caused by the fact that coarser material in the C horizon (upper boundary at 45 cm) reduced the vertical transport rate (k_v in Fig. 1), forcing an accumulation of WEOC (k_s in Fig. 1) directly above this horizon.

The finding that the WEOC below 50 cm was isotropic is certainly of interest. The structure of the variance as a function of tillage direction is presumably related in some way to root activity. However, significant root penetration of the winter wheat crop of 1989 to depths below 75 cm seems unlikely. Köhnlein and Vetter (1953) concluded that only 1% of the root mass of most crops in normal agricultural soil in Fed. Rep. Germany penetrates below 45 cm. This would preclude any significant direct input from the rhizosphere to the WEOC in the deeper horizons. The fact that the WEOC above 50 cm is anisotropic is not surprising, since the mat-like growth of most crop roots at shallower depths (Köhnlein and Vetter, 1953) would tend to obliterate any pattern. It is possible, however, to postulate that a few roots do indeed penetrate deep enough, and may serve as channels of preferential flow, resulting in an accumulation of WEOC through the mobile organic material. If this is the case, it could have a bearing on our understanding of the transport of natural and anthropogenic material to groundwater. Another possibility is that the roots as they draw water into the plants cause an enrichment of organic solutes under the plants. A similar process can effect the distribution of soluble inorganic salts (Anonymous, 1969).

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